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General decision parameters account for shared behavioural and neural variation across interference and response inhibition

Abstract

Interference inhibition and response inhibition correlate only weakly in behaviour, yet recruit overlapping brain regions, which is often taken as evidence for a shared inhibitory mechanism. Group-level overlap, however, cannot reveal whether individual differences in these regions reflect inhibition or the general decision processes that both tasks require. We addressed this question in 106 young adults (M_{age}=22.01 years, *SD* = 2.90; 48 men) who performed *Stroop* and *Go/No-Go* tasks during functional magnetic resonance imaging. Hierarchical drift-diffusion modelling separated general decision components (baseline drift rate, boundary separation and non-decision time) from condition-specific components (incongruent and No-Go drift rates, residualised on baseline drift). We found that all three general parameters correlated across tasks, whereas the condition-specific components did not (r=−.089); an equivalence test rejected correlations of $|r| \geq .30$ but not smaller ones. In meta-analytically defined shared regions, activation in the inhibition contrasts (incongruent > congruent; No-Go > Go) was associated with a composite baseline drift rate, most consistently in the supplementary motor area, and condition-specific components explained no additional variance. Once general components were controlled, each condition-specific component was related to activation in its own task-preferred region: the left intraparietal sulcus for *Stroop* and the right inferior frontal gyrus/premotor cortex for *Go/No-Go*. Together, these findings suggest that the unity of inhibitory control lies largely in general decision processes and its diversity in task-specific computations. Group-level co-activation therefore cannot, by itself, establish a shared inhibitory mechanism, and subtraction contrasts do not necessarily remove general decision

processing from individual differences.

Keywords: inhibitory control; executive functions; drift-diffusion model; individual differences; functional magnetic resonance imaging

1 Introduction

Interference inhibition and response inhibition are two major forms of inhibitory control. The first, typically measured with *Stroop* or *flanker* tasks, resists irrelevant information that competes with task-relevant input; the second, typically measured with *stop-signal* or *Go/No-Go* tasks, withholds or cancels a prepared motor response (Diamond, 2013; Friedman & Robbins, 2022). Behavioural correlations between the two are weak, and the limited shared variance makes a unitary inhibition factor difficult to extract (Rey-Mermet et al., 2018). In the brain, however, meta-analyses identify overlapping activation, particularly in the anterior insula. The extent of overlap in medial frontal regions, including the supplementary motor area (SMA), varies across tasks and analyses (Cieslik et al., 2015; Hung et al., 2018; Isherwood et al., 2021; Sebastian et al., 2013). What does this overlap tell us about the structure of inhibitory control?

Whether interference and response inhibition draw on a common ability determines how inhibition fits within the unity-and-diversity framework of executive functions (Friedman & Miyake, 2017; Miyake et al., 2000). Early latent-variable work suggested that prepotent response inhibition and resistance to distractor interference were closely related (Friedman & Miyake, 2004), and the neural overlap is often read as further evidence for a shared inhibitory ability. The interpretation of neural overlap therefore bears on whether “inhibition” identifies a coherent dimension of individual differences.

Behavioural and neural evidence address different levels of analysis, and the degree of neural overlap depends on the level assessed (Isherwood et al., 2023; Palmer et al., 2025; Swick et al., 2011). Behavioural dissociation is a claim about individual differences; neural overlap is usually a claim about group averages. Shared activation therefore need not imply shared individual differences, and inferring a common cognitive process from it requires additional argument (Henson, 2005; Poldrack, 2006). Shared regional activation can coexist with considerable behavioural independence

across conflict tasks (Fan et al., 2003). More generally, between- and within-person brain–behaviour relationships can run in opposite directions (Mistry et al., 2026). In a sample of 546 participants, for example, a battery of executive function tasks jointly activated canonical control regions, yet activation in those regions was not consistently associated with individual differences in common executive function (Reineberg et al., 2022). Wager et al. (2005) reported a related pattern: *Go/No-Go*, *flanker* and stimulus–response compatibility tasks jointly activated regions including the bilateral anterior insula, which showed coherent individual differences within each task even though cross-task correlations in both activation and behaviour were low. The authors attributed this to a common mechanism for detecting or resolving interference, masked in behaviour by task-specific sources of variance. It remains unclear whether individual differences in shared regions reflect a common inhibitory mechanism or other processes required by both tasks.

Three accounts explain this pattern differently. The shared-inhibition account encompasses several proposed mechanisms. Wager et al. (2005) proposed a common mechanism for detecting or resolving interference, and Wessel and Anderson (2024) proposed a fronto-subthalamic inhibitory mechanism that may operate across domains. On this view, behavioural correlations are weak because standard measures conflate several processing stages, are unreliable and are contaminated by trial-level noise (Draheim et al., 2019; Hedge et al., 2018; Rouder et al., 2023). A general-efficiency account holds instead that the variance shared by inhibition tasks reflects the task-general efficiency of evidence accumulation rather than inhibition per se (Weigard & Sripada, 2021). Consistent with this view, conflict-specific parameters are nearly uncorrelated across conflict tasks, while parameters indexing processing speed and response strategy correlate positively across tasks (Hedge et al., 2022). Within the same sample, a task-general efficiency factor is more cohesive and temporally stable than a control factor built from difference scores (Weigard et al., 2021). In another latent-variable study, the common factor of executive functions could be accounted for by the

speed of information uptake (Löffler et al., 2024). Evidence-accumulation efficiency has also been proposed as a transdiagnostic vulnerability factor (Sripada & Weigard, 2021). Finally, according to a dissociation account, the two forms of inhibition rely at least partly on distinct processes. This view is supported by differences in regional activation between the two task types, by the absence of a clear relationship between model parameters estimated from *stop-signal* and interference tasks, and by differences in the spatial distribution of group-level activation when the same participants perform both (Cieslik et al., 2015; Isherwood et al., 2023).

Distinguishing these accounts requires separating performance into components that can be compared across tasks. The drift-diffusion model provides such a decomposition, separating drift rate (the efficiency of evidence accumulation), boundary separation (response caution), starting-point bias and non-decision time (Forstmann et al., 2016; Ratcliff & McKoon, 2008). Hierarchical Bayesian estimation uses group-level information to constrain individual estimates (Wiecki et al., 2013), and implicit-boundary variants accommodate *Go/No-Go* data, in which withheld responses yield no reaction time (Gomez et al., 2007; Ratcliff et al., 2018). This framework separates general components (baseline drift rate, boundary separation and non-decision time) from condition-specific accumulation on *Stroop*-incongruent trials and for No-Go stimuli. Single-task studies already suggest that inhibition-related activation carries a general component. In the *stop-signal* task, left anterior insula activation correlated with overall accuracy (Boehler et al., 2010), and Go-trial drift rate correlated not only with right frontopolar activation on Go and stop trials but also with stop-trial activation in canonical inhibition regions, including the right inferior frontal gyrus, medial frontal gyrus and basal ganglia (White et al., 2014). In a large *stop-signal* sample, general decision parameters were robustly related to functional connectivity and to cigarette and cannabis use, but inhibition parameters were only weakly related to connectivity (Weng et al., 2026).

The unresolved question is which components account for individual differences

in shared-region activation. Wager et al. (2005) read performance-related activation in shared regions as the signature of a common interference mechanism; single-task modelling studies, by contrast, imply that general decision processes contribute to the same brain–behaviour relationships. Isherwood et al. (2023), who combined a *stop-signal* task, a *multi-source interference* task and computational modelling, found no clear cross-task parameter associations and only limited spatial overlap within participants, but did not ask how general and condition-specific components each contribute to activation in shared regions. Three questions therefore remain open: how much of the individual variation in shared-region activation is attributable to general versus condition-specific components; whether these brain–parameter relationships generalise across tasks; and whether task-specific signals survive once general components are controlled and, if so, where they reside. On the last point, Reineberg et al. (2022) proposed that individual differences in executive function are expressed mainly in domain-specific target regions rather than in shared control regions—a proposal that yields a testable spatial prediction.

Here, the same participants performed a colour–word *Stroop* task and a *Go/No-Go* task during functional magnetic resonance imaging (fMRI); resting-state data were also acquired. We used hierarchical drift-diffusion modelling (HDDM) to separate general from condition-specific components in each task, related these components to activation in shared and task-preferred regions defined by an independent meta-analysis (Isherwood et al., 2021), and tested whether brain–parameter relationships transferred across tasks (Figure 1). The three accounts make distinct predictions about these associations. On the shared-inhibition account, condition-specific components should correlate across tasks and explain activation in shared regions—the SMA and bilateral anterior insula—over and above general components. On the general-efficiency account, cross-task commonality should reside mainly in general decision parameters, which should also account for shared-region activation. On the dissociation account, condition-specific components should be unrelated across tasks and relate mainly to

each task's preferred region as defined by the meta-analysis: the left intraparietal sulcus (IPS) for interference inhibition and the right inferior frontal gyrus/premotor cortex (rIFG/PMC) for response inhibition. The last two accounts are compatible: one identifies the source of shared variance, whereas the other describes inhibition-specific processing. If both hold, general components should explain individual differences in shared regions, whereas condition-specific components should contribute mainly in their own task's preferred region. Together, these predictions address whether inhibition constitutes a unitary dimension of individual differences, and whether neural overlap can count as evidence for such unity.

2 Method

2.1 Participants

Sample size was determined a priori using G*Power 3.1 (Faul et al., 2009). Following recommendations to justify the smallest effect size of interest (Lakens, 2022), we selected $|r| = .30$, which also served as the primary bound for equivalence testing. Detecting a correlation of this size with 80% power (two-tailed $\alpha = .05$) requires 85 participants; allowing for approximately 20% attrition at imaging quality control, we aimed to recruit 110. A sensitivity analysis showed that the final sample of 106 provided 80% power to detect correlations of $|r| \geq .27$.

One hundred and ten healthy adults aged 18–35 years who met the inclusion criteria (Supplementary Methods S1) were recruited between March and July 2026. All completed both tasks, and none was excluded on the basis of task performance. Four were excluded at imaging quality control: two for excessive head motion or elevated standardised spatial root mean square of temporal signal differences (DVARS), and two for incomplete brain coverage in a *Go/No-Go* run. The final sample therefore comprised 106 participants (Table 1). Participants' sex was self-reported. The study was approved by the Ethics Committee of **[Blinded for review] (approval no. [blinded])** and conducted in accordance with the Declaration of Helsinki. All participants provided

written informed consent and were compensated for their time.

2.2 Tasks

Interference inhibition was measured with a colour–word *Stroop* task and response inhibition with a *Go/No-Go* task (MacLeod, 1991; Simmonds et al., 2008). Both tasks used event-related designs, allowing the haemodynamic response to each trial type to be estimated separately (Dale & Buckner, 1997; Garavan et al., 1999), and both were programmed in E-Prime 3.0 (Psychology Software Tools, Inc., 2016).

Stroop task: Participants responded to the ink colour of each stimulus while ignoring its meaning (Stroop, 1935). Stimuli were the Chinese characters for "red" (红) and "green" (绿), printed in red or green; participants responded with the right index finger to red and the right middle finger to green. Congruent, incongruent and neutral trials (36 each; 108 in total) were presented in pseudorandom order; on neutral trials, a symbol string ("###") appeared in red or green. Each stimulus remained on screen for 1,000 ms, during which responses were recorded, and was followed by a fixation cross; inter-trial intervals were jittered between 1,000 and 2,000 ms.

Go/No-Go task: Participants pressed a key with the right index finger as quickly as possible when the Go letter appeared and withheld responding when the No-Go letter appeared. The letters X and Y served as stimuli, with their Go and No-Go roles reversed across the two runs. Each run contained 75 Go and 25 No-Go trials in pseudorandom order. The higher proportion of Go trials was intended to encourage a prepotent Go response (Wessel, 2018). Each stimulus was presented for 500 ms, during which responses were recorded; inter-trial intervals were 1, 3 or 5 s.

2.3 Procedure

Participants practised both tasks outside the scanner and proceeded to scanning once they understood the instructions. The resting-state scan preceded the tasks to avoid carry-over of task-evoked activity (Barnes et al., 2009).

2.4 Image acquisition

Images were acquired on a 3 T United Imaging uMR NX scanner with a 64-

channel head coil. To improve blood oxygen level-dependent (BOLD) signal estimation, all functional runs used a four-echo gradient-echo echo-planar imaging (EPI) sequence with whole-brain coverage (repetition time [TR] = 2,000 ms; 2.5 mm isotropic voxels; Kundu et al., 2017). The resting-state run (eyes open, central fixation) and each of the two *Go/No-Go* runs comprised 200 volumes (≈6.7 min), and the *Stroop* run comprised 139 volumes (≈4.6 min). A T1-weighted structural image was acquired at 0.8 mm isotropic resolution. Full acquisition parameters are given in Supplementary Methods S2.

2.5 Image preprocessing and quality control

Data were preprocessed with fMRIPrep 24.1.1 (Esteban et al., 2019), and the echoes were optimally combined and denoised with multi-echo independent component analysis in tedana 26.0.3 (DuPre et al., 2021; Supplementary Methods S3). All analyses were conducted in MNI152NLin2009cAsym space at 2 mm isotropic resolution. Task data were smoothed with a 6 mm full-width at half-maximum kernel; pattern similarity analyses used unsmoothed data. Resting-state region-of-interest (ROI) time series were detrended, band-pass filtered (0.01–0.08 Hz) and cleaned of head-motion, white-matter, cerebrospinal-fluid and spike confounds in a single regression step (Hallquist et al., 2013); global signal regression (GSR) was applied only in sensitivity analyses. Participants were included in resting-state analyses only if mean framewise displacement (FD) was ≤ 0.30 mm and no more than 20% of volumes exceeded 0.5 mm FD, and were excluded from group-level task analyses if more than 20% of volumes in any run exceeded 0.5 mm FD (Supplementary Figure S1).

2.6 Data analysis

2.6.1 Overview

Analyses were organised into three tiers. Primary analyses, which directly adjudicated among the three accounts, comprised (a) cross-task correlations of general and condition-specific components and (b) hierarchical regressions relating these components to activation in five ROIs; both focused on whether condition-specific

components correlated across tasks and explained variance beyond general components. Secondary analyses comprised variance partitioning, cross-task prediction of shared-region activation from condition-specific components, spatial overlap of activation, cross-task pattern similarity and resting-state connectivity. Cross-validated transfer across tasks was treated as exploratory. All frequentist tests were two-tailed ($\alpha = .05$), except the one-tailed test of cross-task pattern similarity. Within each family of tests, *p* values were corrected with the Benjamini–Hochberg false discovery rate (FDR) procedure and are reported as *q* (Benjamini & Hochberg, 1995). Between-participant analyses controlled for age, sex and years of education; because head motion can confound associations between functional connectivity and behaviour (Siegel et al., 2017), imaging analyses additionally controlled for mean FD in the relevant scan.

2.6.2 Behavioural analyses

For the *Stroop* task, we computed accuracy and mean correct reaction time (RT) for each condition. Given the sensitivity of RT summaries to extreme values (Whelan, 2008), RT analyses included only correct trials within ±3 *SD* of the mean RT. The RT interference effect (incongruent minus congruent) was tested with a paired-samples *t* test (Lakens, 2013). For the *Go/No-Go* task, we computed Go accuracy, correct Go RT and No-Go accuracy, combining the two runs weighted by the number of valid trials; paired-samples *t* tests compared Go with No-Go accuracy and performance across runs. Following Rey-Mermet et al. (2018), Pearson correlations assessed cross-task relations among conventional behavioural indices, with FDR correction across all 16 correlations. Internal consistency was estimated with stratified random split-halves and Spearman–Brown correction (Parsons et al., 2019; Supplementary Methods S4).

2.6.3 Drift-diffusion modelling

We fitted the trial-level data from each task separately using HDDM software (version 1.0.1; Wiecki et al., 2013). The *Stroop* model was stimulus-coded, with the upper and lower boundaries corresponding to the two colour responses, and included all trials with a response so that the RT distributions of both correct and error responses

informed the fit (Ratcliff & McKoon, 2008). Drift rate was estimated separately for congruent, neutral and incongruent trials; the starting point was fixed midway between the boundaries, and boundary separation and non-decision time were shared across conditions. The *Stroop* conflict effect was thus expressed as differences in drift rate between conditions. Because withheld responses on No-Go trials yield no RT, the *Go/No-Go* task was fitted with an implicit-boundary model (Gomez et al., 2007; Ratcliff et al., 2018), in which the upper boundary corresponds to a Go response and the lower boundary to withholding; drift rate was estimated separately for Go and No-Go stimuli. To capture the response bias induced by the 75% Go proportion, the model included a starting-point parameter *z*, shared across conditions (Ratcliff et al., 2018; Weigard et al., 2020). In both models, the proportion of contaminant trials was set to 5%. We fixed inter-trial variability in drift rate and starting point at zero and estimated inter-trial variability in non-decision time at the group level only (Lerche & Voss, 2016). Convergence was assessed with trace plots, the Gelman–Rubin statistic ($\hat{R} < 1.01$) and effective sample size (ESS > 400; Gelman et al., 2013; Vehtari et al., 2021), and model adequacy with 500 posterior predictive simulations of accuracy and RT distributions (Supplementary Methods S5). Condition differences were summarised by the 95% highest-density interval (HDI) of the group-level posterior difference and evaluated against a region of practical equivalence (ROPE; [−0.2, 0.2] for drift rate; Kruschke, 2018).

2.6.4 General and condition-specific components

Individual parameter estimates were taken as posterior means. General components were baseline drift rate (congruent trials for *Stroop*; Go stimuli for *Go/No-Go*), boundary separation and non-decision time. Because drift rates for different conditions within a task are typically highly correlated (Weigard et al., 2020), condition-specific components were defined by residualisation, which makes them linearly independent of baseline drift by construction (DeGutis et al., 2013). The *Stroop* condition-specific component was the standardised residual of incongruent drift

regressed on congruent drift; the *Go/No-Go* condition-specific component was the standardised residual of No-Go drift regressed on Go drift. *Go/No-Go* starting-point bias served as a supplementary index. Hereafter, “general components” denotes baseline drift rate, boundary separation and non-decision time, whereas “specific components” denotes the two residualised effects; “processes” refers to the cognitive operations they are assumed to index.

To index the general processing shared by the two tasks with a single variable in brain–parameter analyses, we standardised each general component within task, averaged the two tasks’ parameters of the same type with equal weights and re-standardised the result. This yielded three composites: shared baseline drift rate, shared boundary separation and shared non-decision time. Drift-rate estimates from different cognitive control tasks load on a common efficiency factor, while response caution and non-decision time also show task-general variance (Hedge et al., 2022; Weigard et al., 2021). Equal weighting emphasises variance common to both tasks and places their general components on a common scale; with only two indicators per parameter, latent-variable estimation was not feasible. To check that the results did not depend on this choice, all brain–parameter analyses were repeated with each task’s own general components in place of the composites (Supplementary Table S1).

Cross-task associations were assessed with partial correlations, separately for general and specific components (Hedge et al., 2022). A paired-participant bootstrap compared the absolute cross-task correlation of each general component with that of the specific components (Efron & Tibshirani, 1993). For the correlation between specific components, we also conducted two one-sided tests of equivalence (TOST), with $|r| = .30$ as the primary bound and .20 and .10 as sensitivity bounds (Lakens et al., 2018), and computed a Jeffreys–Zellner–Siow Bayes factor to quantify evidence for a null correlation (Lee & Wagenmakers, 2014; Liang et al., 2008; Rouder & Morey, 2012; Wetzels & Wagenmakers, 2012). Finally, partial correlations tested whether each parameter was associated with the corresponding within-task behaviour, and

coefficients for task-matched and non-matched parameters were compared directly (Supplementary Methods S5).

2.6.5 Task fMRI analyses

Subject-level general linear models were estimated in Nilearn 0.13.1 (Nilearn contributors, 2026). The *Stroop* model included separate regressors for correct congruent, neutral and incongruent trials and for errors; the primary contrast was correct incongruent minus correct congruent. The *Go/No-Go* model included separate regressors for correct Go, correct No-Go, omission and commission trials; the primary contrast was correct No-Go minus correct Go, combined across runs in a fixed-effects model (Beckmann et al., 2003). Group-level inference used two-tailed nonparametric permutation tests with a cluster-forming threshold of $p < .001$ and cluster-level family-wise error (FWE) correction at $p < .05$ (Eklund et al., 2016; Winkler et al., 2014). Group-level overlap was described as the intersection of the two tasks' separately thresholded positive maps and was not treated as a formal conjunction test (Figure 2B). Subject-level overlap was quantified with the Dice coefficient (Dice, 1945; Supplementary Figure S2; Supplementary Table S2).

To avoid circularity (Kriegeskorte et al., 2009), ROIs were defined from the cluster masks of an independent activation likelihood estimation (ALE) meta-analysis (Isherwood et al., 2021). Shared ROIs were the intersections of the interference- and response-inhibition clusters: the SMA, right anterior insula/frontal operculum and left anterior insula. Task-preferred ROIs were the left IPS, preferentially engaged by interference inhibition, and the rIFG/PMC, preferentially engaged by response inhibition (Supplementary Table S3). For each ROI and contrast, we extracted the mean voxel-wise *t* value to test group-level activation and between-task differences, and tested the ROI × task interaction with a linear mixed model including random intercepts for participants.

To test whether shared regions were engaged with similar spatial patterns across tasks, we correlated each participant's unsmoothed voxel-wise *t* maps for the two

primary contrasts within each shared ROI and Fisher *z*-transformed the result (Kriegeskorte et al., 2008; Palmer et al., 2025). A one-tailed sign-flipping test assessed whether mean similarity exceeded zero (Supplementary Methods S6).

2.6.6 Resting-state functional connectivity

To test whether the task results were mirrored in intrinsic functional organisation, we extracted the confound-regressed, filtered mean time series of each of the five ROIs and computed Fisher *z*-transformed Pearson correlations for all 10 pairs. Guided by the regional structure identified in the independent meta-analysis (Isherwood et al., 2021), we summarised these into four indices: shared-core connectivity, the mean Fisher *z* value across the three connections among the shared regions; *Stroop*-priority connectivity, the mean across the three connections between the left IPS and the shared regions; *Go/No-Go*-priority connectivity, the corresponding mean for the rIFG/PMC; and bridge connectivity, the connection between the two task-preferred regions (Supplementary Methods S7).

2.6.7 Relating model parameters to brain function

Component structure of task activation. Following previous work relating diffusion-model parameters to activation through regression (Kühn et al., 2011; White et al., 2014), we entered standardised mean voxel-wise *t* values for each task and ROI as dependent variables in hierarchical regressions. Step 1 included the covariates and the three composites; Step 2 added the two specific components. The key tests were the increment in explained variance at Step 2 (ΔR^2) and the standardised coefficients of the specific components. In shared ROIs, we compared the coefficients of shared baseline drift rate across tasks to test whether general components played a consistent role. In task-preferred ROIs, we compared the coefficients of the matched and non-matched specific components to test the relative task selectivity of the associations. Coefficients and incremental tests were FDR-corrected separately.

Variance partitioning and cross-task explanation. In shared ROIs, explained variance was partitioned into portions unique to general components, unique to specific

components and common to both (Nimon & Oswald, 2013). As a further test of the shared-inhibition account, we regressed shared-ROI activation in each task on the other task's specific component, controlling for the first task's general components and the covariates, and evaluated null results with equivalence tests and Bayes factors. We also tested whether cross-task pattern similarity was related to the computational components.

Resting-state connectivity. Following work linking resting-state networks to general and specific individual differences in executive function (Reineberg et al., 2015), we regressed each of the four connectivity indices on the three composites, the two specific components and the covariates. To test task selectivity, we compared the coefficients of the matched and non-matched specific components for the two indices linking a task-preferred region to the shared regions, and assessed the joint matching pattern with the sum of the two matched-minus-non-matched contrasts (Supplementary Methods S7).

Cross-validated prediction. To test whether the mapping between general components and shared-region activation generalised across tasks, we trained ridge regression models (Hoerl & Kennard, 1970) on one task's general components and the covariates, then applied them to the other task's parameters of the same type to predict that task's shared-ROI activation. Predictive gain was quantified as the increase in out-of-fold R^2 relative to a covariates-only model. Analyses used nested participant-level cross-validation, with all standardisation and hyperparameter selection confined to the training folds (Poldrack et al., 2020; Varoquaux et al., 2017; Supplementary Methods S8).

Robustness analyses. The main analyses were repeated in a strict head-motion subsample, with GSR, with each task's own general components in place of the composites, and with *Go/No-Go* starting-point bias added as a predictor. For comparison with the computational decomposition, the brain–behaviour analyses were also repeated with conventional behavioural indices (Supplementary Methods S9).

3 Results

3.1 Both tasks produced robust effects, but conventional inhibition indices were unrelated across tasks

Both tasks produced the expected group-level effects. *Stroop* RTs showed an interference effect of 47.50 ms (*SD* = 27.74), $t(105) = 17.63$, $p < .001$, and No-Go accuracy was lower than Go accuracy, $t(105) = -4.16$, $p < .001$. No difference between the two *Go/No-Go* runs survived FDR correction (Figure 3A). Split-half reliability was high for correct RT measures across both tasks (.930–.987) and for Go and No-Go accuracy (.874 and .764), and was .698 for the RT interference effect (Supplementary Table S4).

Across tasks, speed measures were related, but inhibition indices were not. Correct RTs in every *Stroop* condition correlated with correct Go RT (*r*s = .420–.497, all $q < .001$), and congruent *Stroop* RT was negatively associated with Go accuracy ($r = -.243$). The RT interference effect showed no association with No-Go accuracy ($r = -.105$, $q = .378$) or with the No-Go − Go accuracy difference ($r = -.140$, $q = .274$; Figure 3B).

3.2 Model parameters were associated with within-task behaviour

Sampling converged for both models ($\hat{R} = 1.002–1.008$; bulk ESS ≥ 1,187; tail ESS ≥ 946; Supplementary Table S5), and posterior predictive checks showed that the models reproduced accuracy and RT distributions in every condition (Supplementary Figure S3). Drift rate differed between conditions in both tasks. The 95% HDI of the incongruent − congruent drift difference was [−1.99, −1.75], and that of the difference in absolute drift rate between Go and No-Go stimuli was [3.29, 3.61]. Both lay entirely outside the ROPE (Figure 3D).

Each parameter was associated with the within-task behaviour it was expected to index. Controlling for age, sex and education, the *Stroop* drift difference (incongruent − congruent) correlated negatively with the RT interference effect ($r = -.589$). Go drift

rate correlated positively with d' (r = .585), and *Go/No-Go* boundary separation correlated negatively with the commission-error rate (r = −.505). Non-decision time correlated strongly with mean RT in each task (r = .945 and .895), and a starting point closer to the Go boundary was associated with more commission errors (r = .461; all q < .05; Figure 3C).

Parameters were more strongly associated with behaviour from the same task than from the other task. In each of the four matched-versus-non-matched comparisons (*Stroop* drift difference with the RT interference effect, Go drift rate with d', *Stroop* non-decision time with mean RT, and *Go/No-Go* non-decision time with Go RT), the task-matched parameter was more strongly related to behaviour than its counterpart from the other task. The 95% bootstrap confidence intervals (CIs) for the matched − non-matched difference excluded zero in every case (Figure 3F).

3.3 General, but not specific, components were shared across tasks

All three general components correlated across tasks, whereas the specific components did not. With age, sex and education controlled, non-decision time showed the strongest cross-task association (r = .535, 95% CI [.396, .654]), followed by baseline drift rate (r = .353, 95% CI [.185, .512]) and boundary separation (r = .229, 95% CI [.002, .446]; Figures 3E and 3G; Table 2). The specific components showed no significant association (r = −.089, 95% CI [−.307, .138]; BF_{01} = 3.35). An equivalence test rejected correlations of $|r| \geq .30$ (p = .014) but not of $|r| \geq .20$ or $|r| \geq .10$.

Only non-decision time showed a significantly stronger absolute cross-task correlation than the specific components ($\Delta|r|$ = .446, q = .002). The corresponding differences for baseline drift rate (q = .085) and boundary separation (q = .396) did not survive correction.

3.4 General components tracked shared regions; specific components tracked task-preferred regions

3.4.1 Group-level activation overlapped across tasks

The two tasks activated partly overlapping networks. The *Stroop* contrast

(incongruent > congruent) activated the SMA, bilateral inferior frontal gyrus, left temporal pole and left inferior parietal lobule, among other regions. The *Go/No-Go* contrast (correct No-Go > correct Go) activated the bilateral insula, SMA, bilateral supramarginal gyrus and left inferior parietal lobule, among others (cluster-level FWE-corrected $p < .05$; Figure 2A; Supplementary Table S6). The two maps overlapped in the SMA, the right inferior frontal gyrus pars opercularis extending into the anterior insula, a region near the left frontal operculum, the left inferior parietal lobule and the precentral gyrus (Figure 2B).

All five ROIs were significantly activated in both tasks ($q < .05$), and all except the left IPS were more strongly activated by *Go/No-Go* than by *Stroop*. This pattern produced a significant ROI × task interaction, $\chi^2(4) = 34.03$, $p < .001$; in the left IPS, the tasks did not differ (Figure 2C; Supplementary Table S7). Voxel-level FWE-corrected maps are shown in Supplementary Figure S4.

3.4.2 Shared-region activation was associated with general components

Baseline drift rate was related to individual differences in shared-region activation. After correction, shared baseline drift rate was positively associated with activation in five of the six combinations of shared region and task: in both tasks for the SMA and the right anterior insula/frontal operculum, and in the left anterior insula for *Go/No-Go* only (Figure 2E). In no shared ROI did the drift coefficient differ significantly between tasks (Supplementary Tables S1 and S8).

General components uniquely explained 3.0%–19.5% of the variance in shared-ROI activation, and adding the specific components produced no significant increment ($\Delta R^2 \leq .022$, $q \geq .754$). Nor did either task's specific component account for shared-ROI activation in the other task (all six $q \geq .845$; $BF_{01} = 2.14–4.65$; Supplementary Figure S5B; Supplementary Tables S9 and S10).

3.4.3 Specific components were associated with activation in task-preferred regions

Each specific component was related to activation in its own task-preferred region. With general components, the other task's specific component and the covariates

controlled, the *Stroop* specific component was positively associated with left IPS activation (β = .296, 95% CI [.100, .491], q = .020). The *Go/No-Go* specific component was negatively related to rIFG/PMC activation (β = −.322, 95% CI [−.523, −.121], q = .015). Because more negative values of the *Go/No-Go* component denote stronger accumulation towards the no-response boundary, this negative relation means that stronger evidence for withholding was accompanied by greater rIFG/PMC activation. In both regions, the specific components explained additional variance (ΔR^2 = .098 and .092, q = .032; Figure 2E–*F*; Supplementary Tables S1 and S9).

These associations were relatively task-selective. The difference between the matched and non-matched coefficients was .413 in the left IPS (95% bootstrap CI [.123, .692]) and −.239 in the rIFG/PMC (95% bootstrap CI [−.449, −.052]; a negative value indicates a stronger negative association for the matched component; Supplementary Table S8).

3.4.4 Mappings between general components and shared-region activation partly transferred across tasks

In three of the six transfer directions, general components improved out-of-fold prediction beyond covariates alone: the SMA from *Go/No-Go* to *Stroop*, and the right anterior insula/frontal operculum and left anterior insula from *Stroop* to *Go/No-Go*. The reverse directions showed no clear gain (Supplementary Figure S6; Supplementary Table S11). Full out-of-fold performance is shown in Supplementary Figure S5D.

3.5 Pattern similarity, resting-state connectivity and robustness analyses

Within individuals, the shared regions were not engaged with similar spatial patterns across tasks. Cross-task multivoxel pattern similarity did not survive correction in any shared ROI, the median subject-level Dice coefficient was 0, and pattern similarity was unrelated to the computational components (Supplementary Figures S2 and S7; Supplementary Table S2).

At rest, connectivity among the ROIs was largely unrelated to the computational components. All connections among the five ROIs exceeded zero, and connectivity

among the shared regions was the strongest (Figure 4A–C). No individual association between connectivity indices and computational components survived correction. One selectivity contrast did: connectivity between the left IPS and the shared regions was more strongly associated with the matched than with the non-matched specific component ($\Delta\beta = .311$, $q = .046$). This difference held with GSR ($q = .040$) but did not survive correction under the strict motion criterion ($q = .052$). Connectivity between the rIFG/PMC and the shared regions showed no such difference (Figure 4E–*F*; Supplementary Tables S12 and S13).

The main task-fMRI findings were robust to alternative specifications. Both task-preferred associations remained significant in the strict head-motion subsample (n = 100–104) and when each task's own general components replaced the composites ($q \leq .023$). The SMA associations with general components in both tasks, the bilateral anterior insula associations in *Go/No-Go* and all three transfer directions were likewise preserved. The one exception was the *Stroop* association in the right anterior insula/frontal operculum, which no longer survived correction under these specifications ($q = .071$ and .074; Supplementary Tables S1 and S11). *Go/No-Go* starting-point bias was unrelated to any imaging measure (Supplementary Figures S5A and S5C).

For comparison, we also related conventional behavioural indices to the imaging measures. During *Go/No-Go*, No-Go accuracy was related to activation in regions in and around the anterior insula after correction (Figure 2D; Supplementary Figure S8). Associations with resting-state connectivity survived correction only with GSR (Figure 4D; Supplementary Table S14).

4 Discussion

We asked whether the individual differences shared by interference and response inhibition reflect a common inhibitory component or general decision processes. General components correlated across tasks, with the strongest association for non-

decision time. Baseline drift rate tracked activation in shared regions, most consistently the SMA. Specific components, by contrast, were unrelated across tasks and made no significant contribution to shared-region activation. Once general components were controlled, however, each specific component was related to activation in its own task-preferred region. These results provide little support for the shared-inhibition account. Taken together, these findings favour the general-efficiency and dissociation accounts: general components carry shared variance, whereas specific components track activation in task-preferred regions.

4.1 The shared behavioural component extends beyond evidence accumulation

All three general components correlated across tasks, but only non-decision time showed a significantly stronger absolute cross-task correlation than the specific components (Table 2). The two tasks therefore appear to share not only the efficiency of evidence accumulation but also processes outside the decision stage, such as stimulus encoding and response execution (Ratcliff & McKoon, 2008). This pattern is consistent with data-driven accounts of shared dimensions of task performance (Eisenberg et al., 2019), and echoes both the cross-task associations of processing-speed and response-strategy parameters in conflict tasks (Hedge et al., 2022) and the link between common executive function and the speed of information uptake (Löffler et al., 2024). If much of the common variance reflects the speed of encoding and execution, theories of executive function unity may need to look to processes more basic than goal maintenance or top-down biasing. Non-decision time showed the strongest cross-task correlation, whereas baseline drift rate was most closely linked to shared-region activation. Different decision components may thus owe their cross-task commonality to different neural bases, and the common variance of inhibition tasks should not be attributed wholesale to evidence-accumulation efficiency.

Conventional behavioural measures showed a parallel pattern: correct RTs correlated across tasks, whereas the *Stroop* interference effect showed no clear relation

to No-Go accuracy or the No-Go − Go accuracy difference. Although adequate for a difference score, the reliability of the interference effect (.698) was lower than that of the Stroop condition-wise RT measures (.930–.975; Supplementary Table S4), which limits the maximum observable cross-task correlation (Hedge et al., 2018).

4.2 General components track shared-region activation

In shared regions, specific components did not significantly improve model fit beyond general components and did not account for activation across tasks; Bayes factors provided weak to moderate support for these null associations (Supplementary Tables S9 and S10). General components therefore help explain why activation in shared regions varies across individuals. This converges with evidence that stop-trial activation covaries with general decision parameters (White et al., 2014), and with a large stop-signal study in which general decision parameters and their associated networks predicted cigarette and cannabis use, whereas inhibition parameters were unrelated to substance use and only weakly related to functional connectivity (Weng et al., 2026).

Why should shared baseline drift rate be associated with contrast activation? One possibility is that individuals who accumulate evidence efficiently modulate task-positive networks more flexibly in line with task demands; in the *N-back* task, such individuals show greater activation under high load and less under low load (Weigard et al., 2024). Another is that response speed affects BOLD amplitude through time on task (Yarkoni et al., 2009); designs that model trial duration explicitly could separate the two. Previous *Go/No-Go* studies have also linked evidence-accumulation parameters to activation during failed inhibition (Letkiewicz et al., 2024; Weigard et al., 2020). Their focus on error-related activation differs from our focus on meta-analytically defined regions where the two forms of inhibition converge.

Shared baseline drift rate was associated with SMA activation in both tasks, and both effects survived all reported robustness checks. The *Stroop* effect in the right anterior insula/frontal operculum was sensitive to the motion criterion and parameter

specification, and the left anterior insula effect emerged only for *Go/No-Go*. Because between-task differences in drift coefficients were not significant, these differences are better read as variation in the strength of evidence than as region-specific task dependence. Cross-task prediction further showed that the mapping between general components and shared-region activation transferred in three of six cases: in the SMA from *Go/No-Go* to *Stroop*, and in both anterior insula regions from *Stroop* to *Go/No-Go* (Supplementary Figure S5D). The general-decision signal in shared regions was thus at least partly common to the two tasks, most robustly in the SMA (Supplementary Tables S1, S8, and S11).

Conventional indices, by contrast, leave the source of such brain–behaviour links ambiguous. In the diffusion framework, No-Go accuracy depends jointly on drift rate, boundary separation and starting-point bias (Ratcliff et al., 2018; Weigard et al., 2020). Given that general components were associated with shared-region activation, the relation between No-Go accuracy and anterior-insula activation in the *Go/No-Go* task (Figure 2D; Supplementary Figure S8) more likely reflects general decision processing than inhibition. Conventional indices can reveal brain–behaviour links; computational decomposition can help to identify their source.

4.3 Specific components track activation in task-preferred regions

Specific components also carried neural signal, but their associations were concentrated in task-preferred regions. With general components controlled, each task's specific component made an independent contribution in its own preferred region; these effects survived strict motion screening and alternative parameterisations and showed relative task selectivity (Supplementary Tables S1, S8, and S9). Previous studies have found that inhibition-related parameters share little variance across tasks or relate only weakly to functional connectivity (Isherwood et al., 2023; Weng et al., 2026). After accounting for general components, these parameters retained task-specific neural associations. This computational pattern accords with the domain-specific target regions proposed by Reineberg et al. (2022). Because both the left IPS

and the rIFG/PMC were activated at the group level in both tasks, this specificity lies in individual differences rather than in whether activation occurs.

The left IPS is consistently activated in meta-analyses of interference tasks (Isherwood et al., 2021), and intraparietal activity tracks evidence accumulation in human perceptual decisions (Kayser et al., 2010). Its relation to the *Stroop* specific component may therefore reflect the selective accumulation of task-relevant evidence under conflict, in line with diffusion models that explain the resolution of interference as a gradual narrowing of attention onto the target (White et al., 2011). The link between the *Go/No-Go* specific component and the rIFG/PMC is consistent with meta-analytic evidence for response-inhibition-related activation in this region (Isherwood et al., 2021). Because this component indexes accumulation towards the no-response boundary independently of Go drift, part of the individual variation in this region appears to be independent of general decision efficiency. At rest, connectivity between the left IPS and the shared regions showed relative selectivity in the same direction (Figure 4F; Supplementary Table S13), providing converging evidence from intrinsic connectivity; this effect, however, did not survive correction under the strict motion criterion, and no corresponding effect emerged for *Go/No-Go*.

4.4 What neural overlap can and cannot show

The computational decomposition also clarifies what neural overlap establishes about shared processing. Group-level co-activation establishes that a region participates in two tasks; it does not establish that individual differences in that region reflect the same cognitive component in both. Inferring a common mechanism from overlap requires additional evidence about the cognitive processes that produce the activation (Henson, 2005; Poldrack, 2006).

This distinction helps resolve the apparent tension between behavioural dissociation and neural overlap in inhibitory control. Group-level overlap is fully compatible with the absence of a cross-task association between specific components, and co-activation alongside relative behavioural independence has been observed

before (Fan et al., 2003). The distinction also helps reconcile our findings with earlier within-participant work. Isherwood et al. (2023) found no clear cross-task association between the model parameters of a *stop-signal* and a multi-source interference task, and only limited spatial overlap within participants. Our findings connect this dissociation to a positive result: despite low cross-task pattern similarity, general components linked the tasks in shared regions. Comparisons across studies should therefore keep three questions apart: whether two tasks activate the same region; whether that region is engaged with similar spatial patterns within individuals; and whether individual differences in that region reflect the same component across tasks.

The same logic applies to subtraction. Contrasts such as incongruent − congruent or No-Go − Go are intended to reduce contributions common to both conditions, but their cognitive interpretation depends on assumptions about which processes differ between conditions (Henson, 2005). In our data, general components remained associated with contrast activation across individuals, showing that subtraction did not eliminate their contribution to individual differences. This parallels the behavioural finding that difference scores can have limited reliability and retain contributions from general processing (Hedge et al., 2018; Weigard et al., 2021), and implies that the validity of subtractive logic must be established separately for group effects and for individual differences. Studies that treat *Stroop* or *Go/No-Go* contrast activation as a neural index of "inhibitory ability" should therefore account for the contribution of general components, as Weigard et al. (2020) argued for *Go/No-Go* activation; inhibition-specific individual differences are more likely to emerge in task-preferred regions once these components are taken into account.

4.5 Implications for the unity and diversity of executive functions

The results identify distinct computational contributions to the unity and diversity of executive functions. First, general decision processes carry much of the shared variance within inhibition. In the unity-and-diversity model, response inhibition is typically indistinguishable from the common executive function factor (Friedman &

Miyake, 2017). At the level of computational components, we found that the common variance of the two inhibition tasks was concentrated in general components such as baseline drift rate and non-decision time, with no evidence of cross-task commonality in specific components. This supports the view that common executive function largely reflects the speed of information uptake or cognitive efficiency (Löffler et al., 2024; Weigard et al., 2021), and extends it from behaviour to the brain regions thought to implement common inhibition.

Second, diversity appears to reflect task-dependent local adjustments rather than separable inhibitory abilities. The specific components index selective accumulation under conflict and accumulation towards the no-response boundary. Individual differences in these components map onto each task's preferred region. This is compatible with network-level challenges to a dedicated inhibitory module (Erika-Florence et al., 2014; Hampshire, 2015) and with the difficulty of extracting a unitary inhibition factor from behaviour (Rey-Mermet et al., 2018). In these two tasks, then, inhibitory control was better described as general decision processes plus task-specific computations than as a single ability that different tasks tap to different degrees.

4.6 Limitations and future directions

Several limitations qualify these conclusions. First, each form of inhibition was measured with a single task in healthy young adults. Because neural mechanisms vary across tasks and task variants (Isherwood et al., 2021; Palmer et al., 2025; Raud et al., 2020), independent replication across additional paradigms and populations is needed to establish generalisability. Second, task design leaves room for refinement. The two-colour *Stroop* task tied each incongruent word to a single correct response, potentially permitting contingency learning (Schmidt & Besner, 2008). Although No-Go accuracy was significantly lower than Go accuracy and starting-point bias was associated with commission errors, the long, variable inter-trial intervals of the *Go/No-Go* task may have reduced response prepotency relative to faster-paced designs (Wessel, 2018). Moreover, because No-Go stimuli were less frequent than Go stimuli, the No-Go − Go

contrast may also capture attentional responses to infrequent stimuli (Hampshire, 2015). Designs that manipulate stimulus frequency independently of response requirements would help distinguish these contributions. Paradigms that manipulate interference and response inhibition within a single task, such as the hybrid response inhibition task (Sebastian et al., 2013), would allow general and specific components to be compared under matched stimulus and response demands. Third, the two tasks required different model structures (two-boundary vs implicit-boundary), and parameters with the same name may combine shared and task-specific variance (Schubert et al., 2016). Within-task standardisation, residualisation and sensitivity analyses using task-specific parameters made the parameters statistically comparable, but a rigorous comparison of constructs would require several tasks per form of inhibition, latent-variable models and tests of measurement invariance (Meredith, 1993; Vandenberg & Lance, 2000). Applying the present decomposition within such a framework would test whether the dissociation observed here holds at the level of latent constructs. Fourth, the sample was powered to detect correlations of $|r| = .30$ or larger. Larger samples would sharpen estimates of smaller brain–behaviour associations (Marek et al., 2022), and brain stimulation or lesion studies could establish the causal contributions of the regions identified here.

5 Conclusion

The individual differences shared by interference and response inhibition were carried mainly by general components, which correlated across tasks and were associated with activation in shared regions, most consistently the SMA. Specific components showed no significant cross-task association and were instead related to activation in each task's preferred region: the left IPS for *Stroop* and the rIFG/PMC for *Go/No-Go*. We found no evidence for a shared inhibition-specific component. The computational decomposition explains how neural overlap can coexist with behavioural dissociation. General components link the tasks in shared regions, while specific

components retain distinct neural associations.

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Table 1. *Participant characteristics*

Characteristic	*n* (%)	*M* (*SD*)	Range
Age (years)	—	22.01 (2.90)	18–32
Sex			
Male	48 (45.3)	—	—
Female	58 (54.7)	—	—
Ethnicity			
Han	96 (90.6)	—	—
Other	10 (9.4)	—	—
Education (years)	—	15.61 (2.41)	12–22
BMI (kg/m²)	—	21.45 (3.10)	16.02–32.74
Socioeconomic status[a]	—	26.69 (6.93)	5–46
Raven's matrices score	—	50.45 (7.65)	0–60

Note. Values are frequencies (percentages) or means (standard deviations). BMI, body mass index.

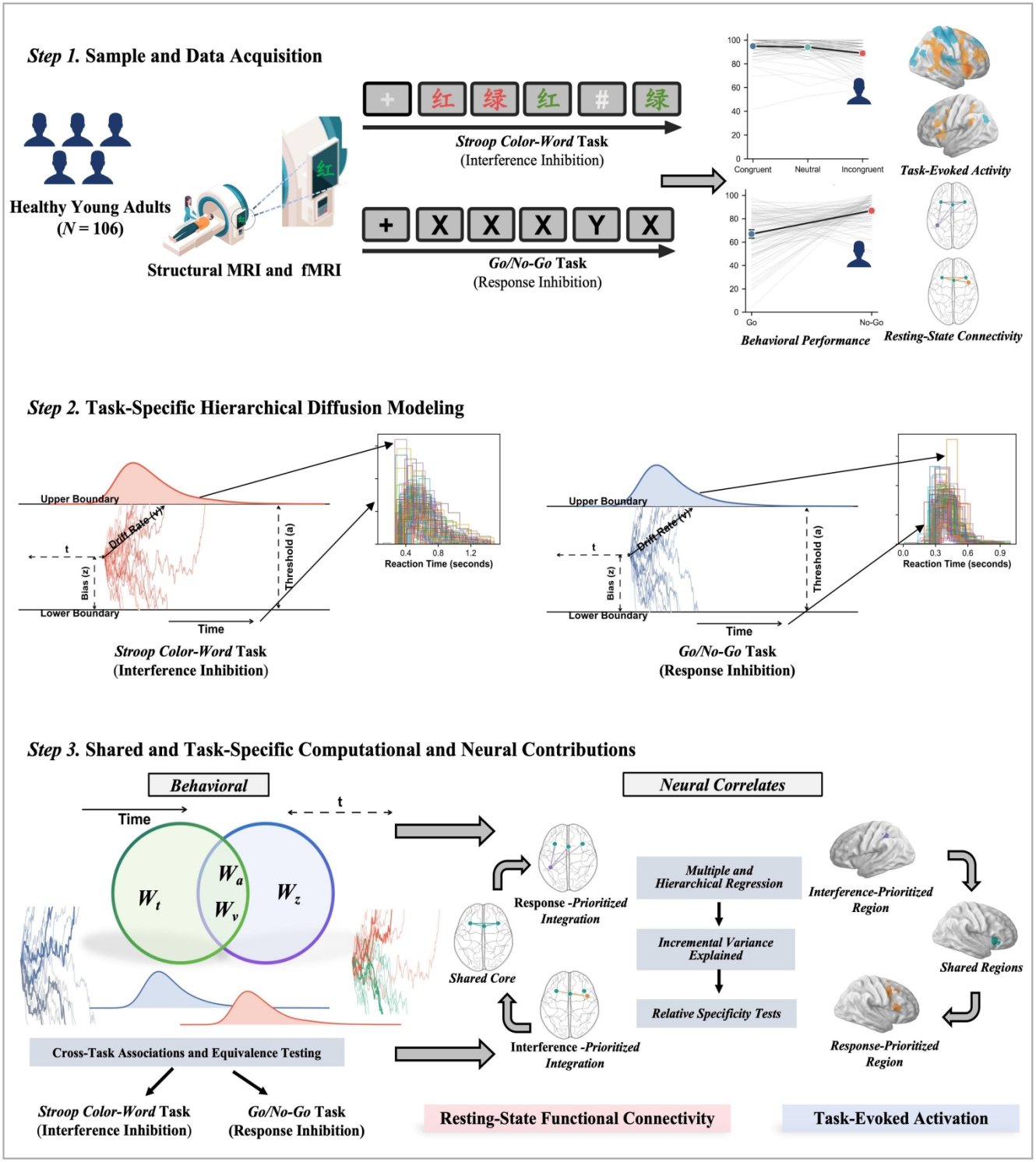
[a] Higher scores indicate higher socioeconomic status; scoring details are provided in Supplementary Methods S1.

Table 2. *Cross-task correlations of general and condition-specific components.*

Component	Partial *r*	95% CI	*p*	*q*	Δ|*r*|	95% CI	Family-adjusted 95% CI	*q*
Baseline drift rate	.353	[.185, .512]	< .001	< .001	.264	[−.009, .445]	[−.059, .483]	.085
Boundary separation	.229	[.002, .446]	.020	.017	.140	[−.157, .369]	[−.223, .421]	.396
Non-decision time	.535	[.396, .654]	< .001	< .001	.446	[.192, .598]	[.135, .630]	.002
Condition-specific components	−.089	[−.307, .138]	.371	.371	—	—	—	—

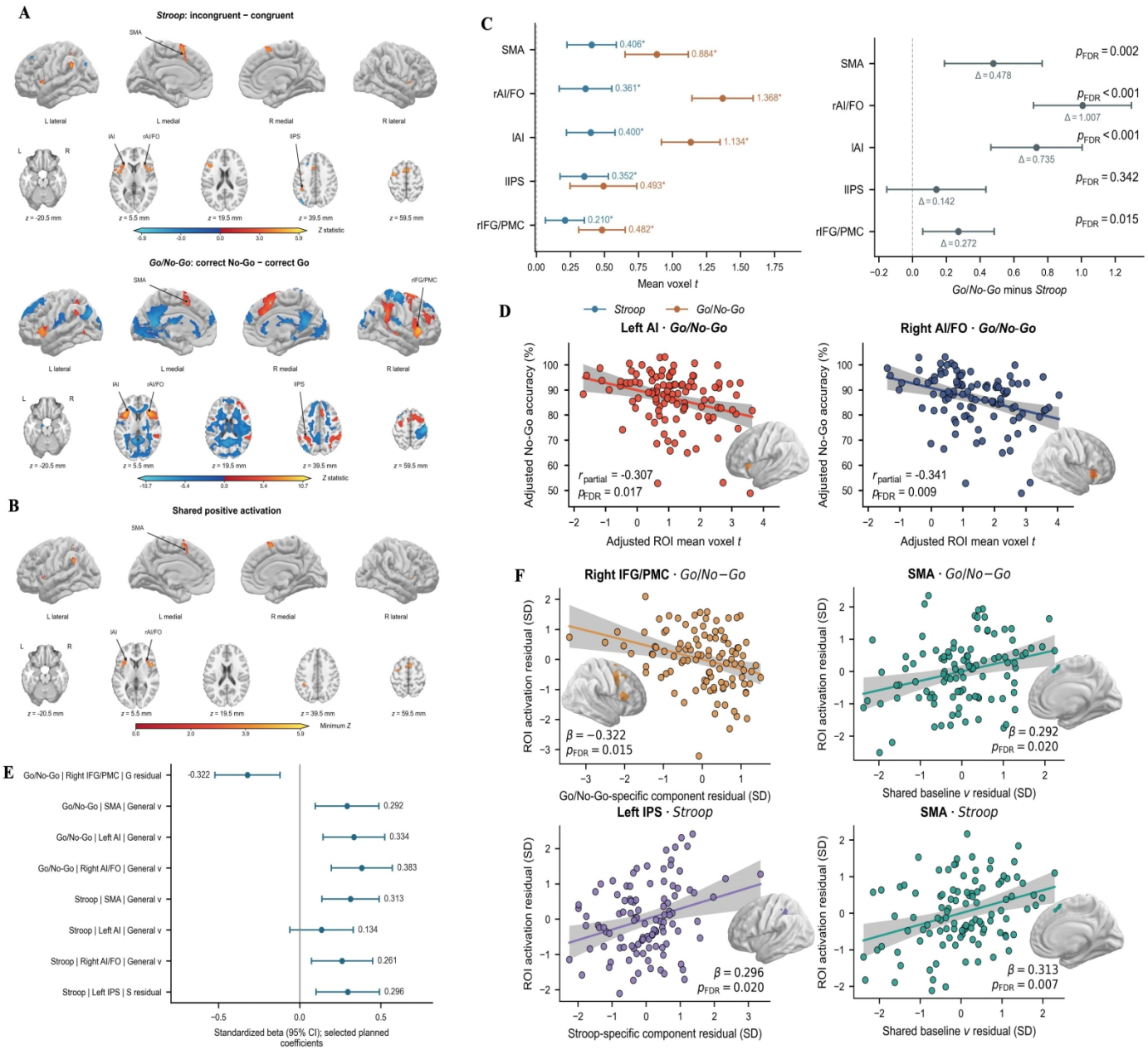
Note. Partial correlations relate each component across tasks, adjusting for age, sex and education. *q* values for the general parameters are corrected across three tests; the condition-specific correlation was tested separately (equivalence test against ±.30: $p = .014$; $BF_{01} = 3.35$). Δ|*r*| is the absolute cross-task correlation of each general parameter minus that of the condition-specific components. Intervals are based on 5,000 participant-level bootstrap resamples, with the condition-specific components refitted in each resample; family-adjusted intervals apply Bonferroni correction across the three comparisons, and the comparison *q* values apply false discovery rate correction within the same family. CI, confidence interval.

Figure 1. *Study design and analytical framework.*



Note. The framework combines behavioural and MRI measures from 106 young adults, task-specific hierarchical drift-diffusion models (HDDM) and cross-task analyses. Regional and network illustrations mark the predefined analysis targets. *v*, drift rate; *a*, boundary separation; *t*, non-decision time; *z*, starting-point bias; MRI, magnetic resonance imaging.

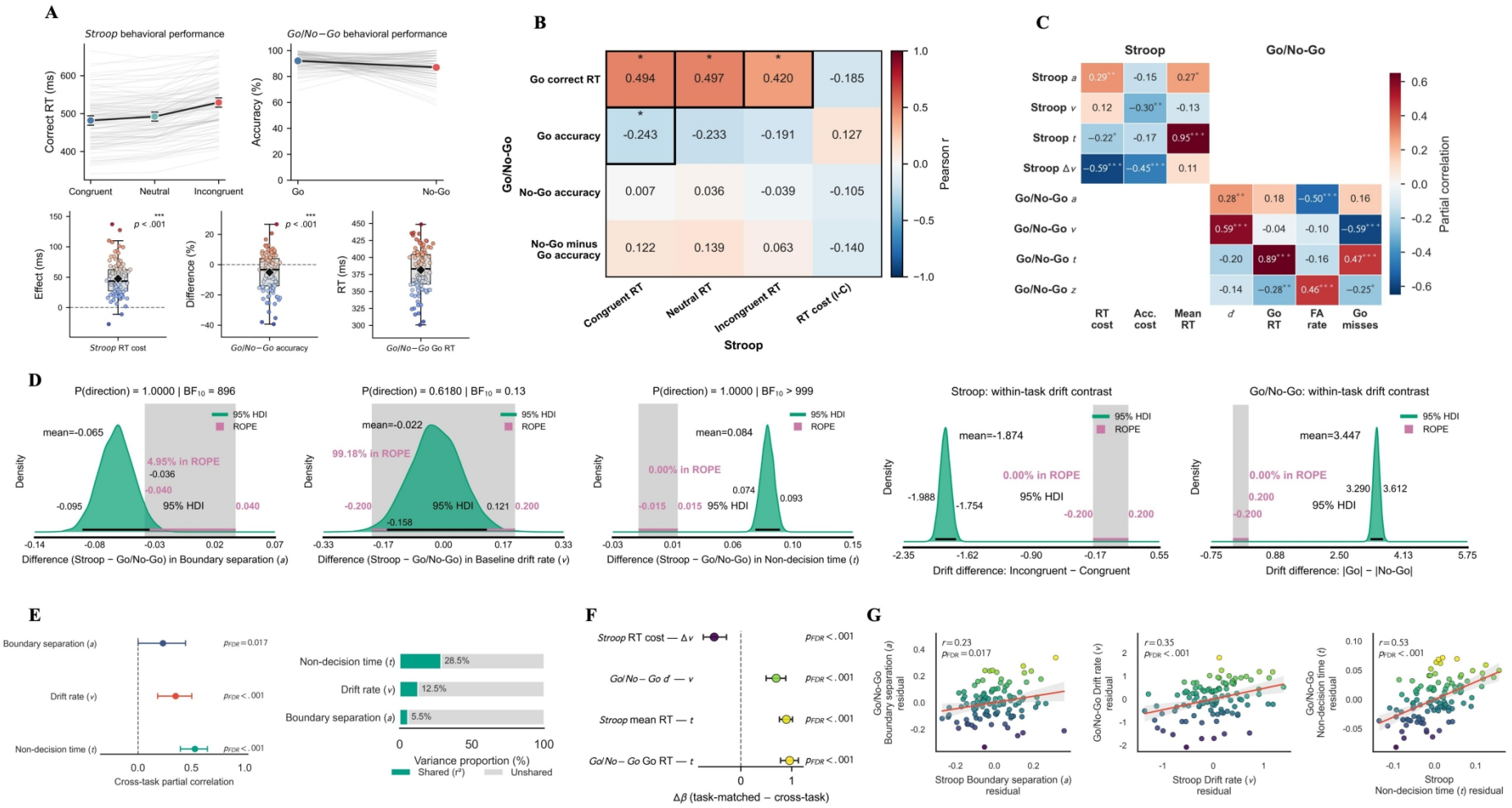
Figure 2. ***Task activation and individual differences.***



Note. (A) *Stroop* incongruent > congruent and *Go/No-Go* correct No-Go > correct Go maps, with cluster-level family-wise error correction at $p < .05$ (cluster-forming $p < .001$). Warm and cool colours indicate positive and negative effects. (B) Descriptive overlap of the positive maps; colours show the smaller of the two task-specific *Z* values. (C) Mean regional voxel-wise *t* values and paired differences with 95% confidence intervals (CIs); asterisks mark activation tests surviving false discovery rate (FDR) correction across 10 tests. (D) Associations between regional activation and conventional behavioural indices. (E) Selected standardised coefficients for shared baseline drift rate and the condition-specific components. (*F*) Corresponding partial regression plots. Lines and shading in D and *F* show fits and 95% CIs for the fitted mean. Associations adjust for age, sex, education and task mean framewise

displacement; computational models include three general and two condition-specific components. FDR correction is applied within each analysis family. ROI, region of interest; SMA, supplementary motor area; AI, anterior insula; FO, frontal operculum; IPS, intraparietal sulcus; IFG, inferior frontal gyrus; PMC, premotor cortex; S residual, *Stroop* condition-specific component; G residual, *Go/No-Go* condition-specific component.

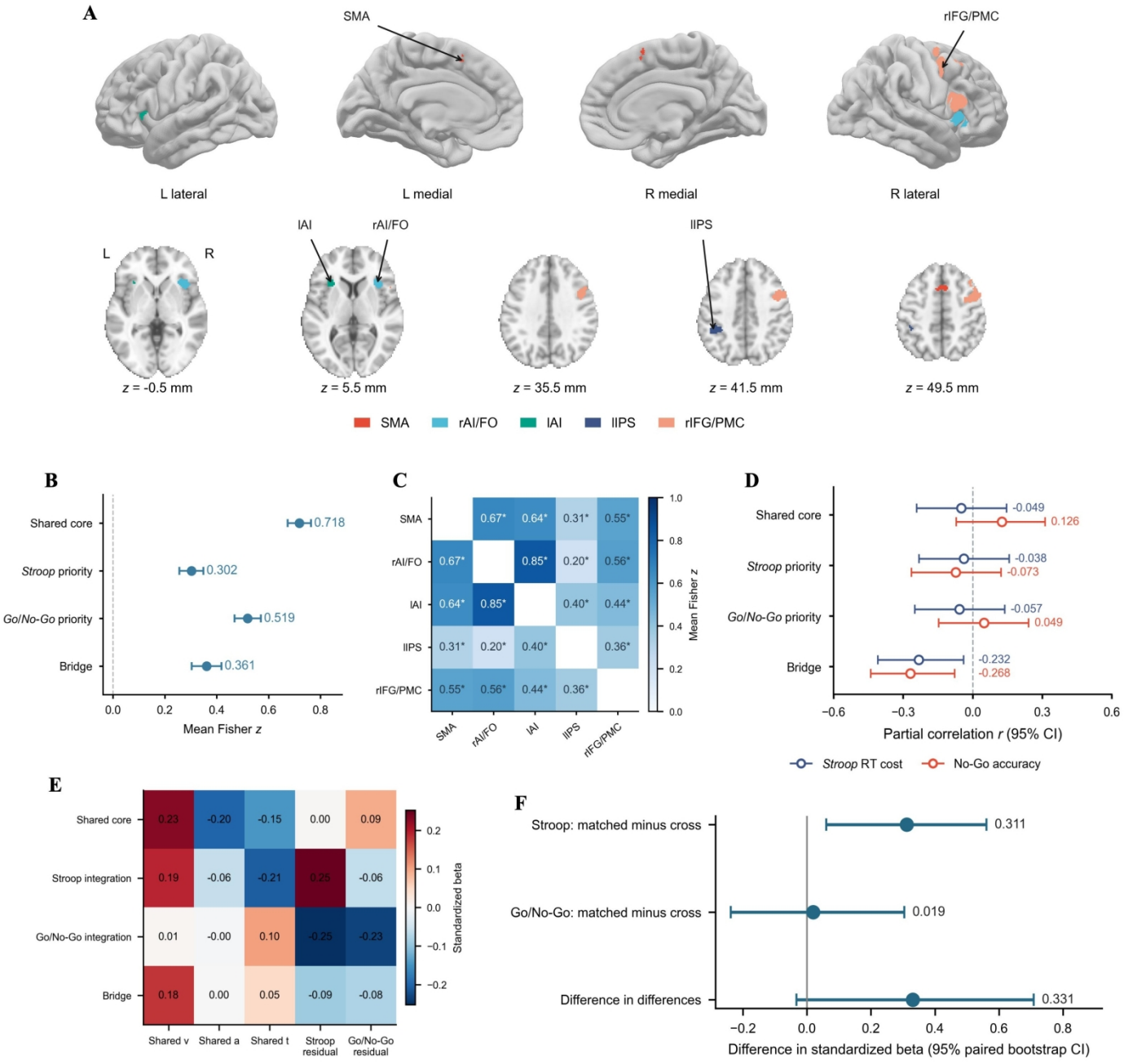
Figure 3. ***Behavioural performance and diffusion-model parameters.***



Note. (A) Behavioural measures. Upper plots show means and 95% confidence intervals (CIs), with paired observations connected; lower plots

show individual observations, means (black diamonds) and interquartile ranges (boxes), with whiskers extending to observations within 1.5 interquartile ranges. (B) Cross-task Pearson correlations; asterisks and outlines denote false discovery rate (FDR)-adjusted $p < .05$ across 16 tests. (C) Parameter–behaviour partial correlations. (D) Cross-task group-level parameter contrasts (*Stroop* minus *Go/No-Go*; three left-hand panels) and within-task drift-rate contrasts (two right-hand panels), with 95% highest-density intervals (HDIs) and regions of practical equivalence (ROPEs). (E) Cross-task partial correlations with 95% participant-bootstrap CIs; bars partition r^2 and $1 - r^2$. (*F*) Task-matched minus non-matched coefficients with 95% participant-bootstrap CIs. (G) Partial-regression scatterplots corresponding to E; lines and shading show fits and 95% CIs. *a*, boundary separation; *v*, drift rate; *t*, non-decision time; *z*, starting-point bias; Δ*v*, *Stroop* drift-rate contrast; RT, reaction time; Acc., accuracy; FA, false alarm (commission error).

Figure 4. *Resting-state connectivity.*



Note. (A) Five regions of interest (ROIs) defined by an independent activation likelihood estimation (ALE) meta-analysis. (B) Four connectivity indices, shown as mean Fisher *z* values with 95% confidence intervals (CIs): shared core, mean connectivity among the three shared ROIs; *Stroop* priority, mean connectivity between the left IPS and the shared ROIs; *Go/No-Go* priority, mean connectivity between the rIFG/PMC and the shared ROIs; and bridge, connectivity between the left IPS and the rIFG/PMC. (C) Mean pairwise connectivity; asterisks denote false discovery rate (FDR)-adjusted $p < .05$ across 10 connections. (D) Partial correlations with *Stroop* reaction-time (RT) interference and No-Go accuracy, with 95% CIs. (E) Standardised coefficients for the general and condition-specific components; the "integration" labels

denote the priority-connectivity indices shown in B and D. (*F*) Task-matched minus non-matched coefficients and their summed contrast, with 95% paired-bootstrap CIs; the point labelled “Difference in differences” denotes the sum of the two matched-minus-non-matched contrasts. D–*F* adjust for age, sex, education and mean resting-state framewise displacement; E–*F* include all general and condition-specific components. L, left; *R*, right; SMA, supplementary motor area; AI, anterior insula; FO, frontal operculum; IPS, intraparietal sulcus; IFG, inferior frontal gyrus; PMC, premotor cortex.